Jose Carvajal-Beltran

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Education

Massachusetts Institute of Technology (MIT) Cambridge, MA
B.S. Materials Science and Engineering (Course 3-A), Physics Concentration. *May 2029*
Relevant Coursework: 6.200 Electrical Circuits: Modelling and Design of Physical Systems, 3.013 Mechanics of Materials, 3.010 Structure of Materials

Experience

Undergraduate Researcher, CNT Growth for BNNT Development Sep. 2026 – Present
MIT UROP, necstlab
- Grow CNT forests via chemical vapor deposition and plasma-etch the arrays to support development of parametric BNNT arrays.
- Measure CNT forest height and mass, maintain structured sample records, and support surface-area and functional-group characterization.

Undergraduate Teaching Assistant & Grader, Physics 8.01 Sep. 2026 – Dec. 2026
MIT Department of Physics
- Support the course's highest-enrollment section in MIT's TEAL format, facilitating collaborative problem-solving and group activities in introductory classical mechanics.
- Provide individualized guidance during class and grade weekly problem sets for accuracy, reasoning, and clear presentation.

Research Intern, Carbon Nanostructure Polymer Composites Jun. 2026 – Aug. 2026
Universidad Anáhuac Mayab, MIT MISTI Program
- Selected through MIT MISTI for a placement under Dr. Andrés I. Oliva-Avilés, contributing to a SECIHTI-funded national research initiative.
- Fabricated carbon nanotube polymer nanocomposites across a weight percent series, tuning ultrasonication parameters and applying electric-field-assisted casting to align filler networks during cure.
- Characterized piezoresistive response across multiple variables and presented findings via literature review.

Undergraduate Researcher, Volumetric Additive Manufacturing Feb. 2026 – May 2026
MIT UROP, Wallin Group
- Optimized azimuthal and polar rotation angles for VAM via spatial frequency analysis on benchmark CAD geometries.
- Extracted orientation heuristics across seven geometries and resolved an inscribed-sphere fitting artifact, reducing print preparation iterations upon codebase integration.

Student Shop Mentor Sep. 2025 – Present
MIT Edgerton Center
- Train peers on shop equipment, manage workspace operations and after-hours access, and advise on fabrication projects.

Projects

Consumer Laser Product Compliance May 2026 – Present
FDA Citizen Petition, Docket FDA-2026-P-7822
- Researched, drafted, and filed a Citizen Petition with the FDA under 21 CFR 10.30 seeking enforcement of accessible emission limits against mislabeled consumer laser pointers.

Mu-Zero Air Hockey Table Jul. 2025 – Sep. 2025
Electronics Lead, MIT Edgerton Center
- Wired a 120V AC OEM blower with safety interlocks and built two beam-break sensor circuits on protoboard for automatic scorekeeping; presented to an audience of 80+, including industry guests.

Autorotation and Reentry Modelling Aug. 2021 – May 2024
Independent Researcher
- Modelled flight dynamics of an autorotating maple samara and extended the principles to a theoretical rotor-equipped reentry capsule; presented twice at ISEF, earning two special awards (CIA + Scholarship).

Leadership

Selected Member, Student Advisory Group in Engineering (SAGE) Oct. 2026 – Present
MIT School of Engineering
- Advise the Dean of Engineering regarding undergraduate academic affairs, research initiatives, and campus life experience.

Skills

Computation: Python (NumPy, SciPy, Matplotlib), Simulation and optimization pipeline design, Git, LaTeX

Laboratory and Fabrication: Ultrasonication and nanofiller dispersion, Electric-field-assisted casting, Piezoresistive and viscoelastic characterization, Optical power measurement, Analog and digital circuits, Arduino, 3D printing